J1
Conn_01x05_Pin
1 SWDIO
2 SWCLK
3 RESET
4
5 3V3_PROT
GND
3V3_PROT
R1
10K
RESET
C3
0.1uF
GND
RESET AND PROGRAMMING CIRCUITRY (SAMD21G18)

C6
22pF
GND
Y2
Crystal
XIN32
XOUT32
C7
22pF
GND
EXTERNAL CRYSTAL (SAMD21G18)

3V3_PROT
C11 10uF
C12 0.1uF
C13 0.1uF
C14 0.1uF
C15 0.1uF
GND
DECOUPLING CAPACITORS (SAMD21G18)

C2
1uF
3V3_PROT
GND
U3
ATSAMD21G18A-A
XIN32
XOUT32
STB
INT
WDI
MISO
MOSI
SCK
CS1
RST_LORA
DIO0
DIO1
CS2
BNO_BOOTN
H_INTN
BNO_NRST
SWCLK
SWDIO
RESET
PA00 PA01 PA02 PA03 PA04 PA05 PA06 PA07 PA08 PA09 PA10 PA11 PA12 PA13 PA14 PA15 PA16 PA17 PA18 PA19 PA20 PA21 PA22 PA23 PA24 PA25 PA27 PA28 PA30 PA31
PB02 PB03 PB08 PB09 PB10 PB11 PB22 PB23
VDDCORE VDDIN VDDIO VDDIO VDDANA GND GNDANA
GND AGND
SAMD21G18

C1
0.1uF
3V3_PROT
GND
U1
MCP2517FD-xSL
C4
22pF
GND
Y1
Crystal
C5
22pF
GND
MOSI MISO CS2 SCK
SDI SDO CS SCK
OSC2 OSC1
VDD VSS
RXCAN TXCAN CLKO/SOF INT INT0/GPIO0/XSTBY INT1/GPIO1
CANRX CANTX INT STB
GND
CAN CONTROLLER

C8
0.1uF
3V3_PROT
GND
U4
TCAN332
CANTX CANRX
TXD RXD VCC GND CANH CANL
CAN_A_H
CAN_A_L
GND
CAN TRANSCEIVER

CAN PERIPHERALS

R3
4.7K
R4
4.7K
3V3_PROT
I2C_A_SDA
I2C_A_SCL

U2
2472
3V3_PROT
BNO_NRST
I2C_A_SDA
I2C_A_SCL
3VO PS0 PS1 INT ADR RST VIN SDA SCL GND
H_INTN
GND
IMU

3V3_PROT
R2
10K
BNO_NRST
SW1
SW_Push
GND
BNO055 RESET CIRCUIT

C9
0.1uF
3V3_PROT
GND
U5
RFM96W-433S2
SCK MOSI MISO NSS RESET
3.3V
ANT DIO5 DIO4 DIO3 DIO2 DIO1 DIO0 GND GND GND
J2
Conn_Coaxial
GND
DIO1
DIO0
LoRa Transceiver
GND
LoRa

SW2
SW_Push
3V3_PROT
R5
10K
RST_LORA
GND
LoRa Reset

C10
0.1uF
3V3_PROT
GND
U6
MAX6373
WDI
3V3_PROT
WDI WDO VCC SET0 SET1 SET2 GND
D1
PMEG4050EP
RESET
GND
WATCHDOG

J3
Conn_01x02_Pin
1 OUT2_MT3
2 OUT1_MT3
J4
Conn_01x02_Pin
1 OUT2_MT2
2 OUT1_MT2
J5
Conn_01x02_Pin
1 OUT2_MT1
2 OUT1_MT1

U7
DRV8830DRCR
R7
R
+3.3V
C16
0.1uF
GND
OUT2_MT3
OUT1_MT3
OUT2 ISENSE OUT1 VCC GND
SCL SDA A1 A0 FAULTN
I2C_A_SCL
I2C_A_SDA
+3.3V

U8
DRV8830DRCR
R10
R
+3.3V
C19
0.1uF
GND
OUT2_MT2
OUT1_MT2
OUT2 ISENSE OUT1 VCC GND
SCL SDA A1 A0 FAULTN
I2C_A_SCL
I2C_A_SDA
+3.3V
GND

U10
DRV8830DRCR
R12
R
+3.3V
C20
0.1uF
GND
OUT2_MT1
OUT1_MT1
OUT2 ISENSE OUT1 VCC GND
SCL SDA A1 A0 FAULTN
I2C_A_SCL
I2C_A_SDA
GND

PC104 connectors
J6
PC104_H1_H2
H1
H2
Vsys SP6 SP5 SP4 SP3 SP2 SP1
PGND
RBF_A DEP_SW_A SP13 SYS_FAULT_n SP11 ENA_5V SP9 PPS SP7 I2C_B_SCL I2C_B_SDA I2C_A_SCL I2C_A_SDA CAN_B_L CAN_B_H DGND CAN_A_L CAN_A_H DGND AGND VREF THM6 THM4 AGND THM2 THM0
RBF_B DEP_SW_B SP14 WDT_KICK SP12 ENA_3V3 SP10 SP8 DGND AGND THM7 THM5 THM3 THM1
notes:
make sure u are using the latest PC104
Please use the same net label neames as used here
SP = spare
DGND GND
PC104

OCPC
VIN(UV) = VUVLO(R) x (Ra + Rb + Rc)/(Rb+Rc)
VIN(OV) = VOVLO(R) x (Ra + Rb + Rc)/Rc
RILIM = 50000/ILIM
Where VUVLO(R) is the EN/UVLO pin rising threshold and VOVLO(R) is the OVLO pin rising threshold.
+3.3V
R6
Ra
UVLO_3V3
R8
Rb
OVLO_3V3
R9
Rc
GND
U9
TPS25961DRVR.A
+3.3V
C17
0.1uF
GND
OVLO_3V3
UVLO_3V3
IN OVLO EN_UVLO
OUT ILIM GND GND
3V3_PROT
R11
RILIM
C18
4.7uF
GND

Sheet: /
File: ADCS_BEACON.kicad_sch
Title:
Size: A3 Date: Rev:
KiCad E.D.A. 10.0.7 Id: 1/1